

$X$  and  $Y$ , with Lagrangian density

$$\mathcal{L}^\# = \frac{1}{2} \left[ \Phi^\dagger e^{-V} \Phi \right]_D + 2 \operatorname{Re} \left[ Y f(\Phi) \right]_{\mathcal{F}} + \frac{1}{2} \operatorname{Re} \left[ X \sum_{A\alpha\beta} \epsilon_{\alpha\beta} W_{A\alpha L} W_{A\beta L} \right]_{\mathcal{F}}. \quad (27.6.4)$$

This Lagrangian density becomes equal to the original one when the scalar components  $x$  and  $y$  of  $X$  and  $Y$  are given the values  $x = 1/g^2$  and  $y = 1$ , and the spinor and auxiliary components of  $X$  and  $Y$  are set equal to zero. Since supersymmetry and gauge invariance are assumed to be preserved in the cut-off procedure, the effective Lagrangian density in the presence of these external superfields must be the sum of the  $D$ -term of a general superfield and the real part of the  $\mathcal{F}$ -term of a left-chiral superfield:

$$\mathcal{L}^\#_\lambda = \left[ \mathcal{A}_\lambda(\Phi, \Phi^\dagger, V, X, X^\dagger, Y, Y^\dagger, \mathcal{D} \cdots) \right]_D + 2 \operatorname{Re} \left[ \mathcal{B}_\lambda(\Phi, W_L, X, Y) \right]_{\mathcal{F}}, \quad (27.6.5)$$

with  $\mathcal{A}_\lambda$  and  $\mathcal{B}_\lambda$  both gauge-invariant functions of the displayed arguments. We do not include any superderivatives or spacetime derivatives in the  $\mathcal{F}$ -term because, as in Section 26.3, terms involving derivatives of any of the left-chiral superfields or their adjoints may be rewritten as contributions to  $[\mathcal{A}_\lambda]_D$ . (It is true that Eq. (27.3.12) gives  $W_L$  itself in terms of two  $\mathcal{D}_{RS}$  acting on a superfield  $\exp(-2V)\mathcal{D}_L \exp(2V)$ , but this superfield is not gauge-invariant, and we are requiring that  $\mathcal{A}_\lambda$  be gauge-invariant.)

The dependence of  $\mathcal{B}_\lambda$  on  $X$  and  $Y$  is severely limited by two additional symmetries of the action obtained from the Lagrangian density (27.6.4). (Both of these symmetries are broken by non-perturbative effects, which will be considered in Chapter 29.) The first symmetry is a perturbative  $U(1)$   $R$ -symmetry, of the sort discussed in Section 26.3, for which  $\theta_L$  and  $\theta_R$  are given the  $R$  values  $+1$  and  $-1$ , the superfields  $\Phi$ ,  $V$ , and  $X$  are  $R$ -neutral, and  $Y$  has the  $R$  value  $+2$ . (Recall that  $f_{\mathcal{F}}$  is the coefficient of  $\theta_L^2$  in  $f$ , so in order for  $f_{\mathcal{F}}$  to have the  $R$  value 0,  $f$  must have the  $R$  value 2.) Because  $W_L$  is given by two  $\mathcal{D}_{RS}$  and one  $\mathcal{D}_L$  acting on  $R$ -neutral superfields, it has the  $R$  value  $+1$ . Now,  $R$  invariance requires  $\mathcal{B}_\lambda$ , like the superpotential, to have the  $R$  value  $+2$ . It cannot depend on any superfields with negative  $R$  values, such as adjoints of left-chiral superfields, because it is *holomorphic*, so the terms in  $\mathcal{B}_\lambda$  can only be of first order in  $Y$  or of second order in  $W_L$ , with coefficients depending only on the  $R$ -neutral superfields  $\Phi$  and/or  $X$ :

$$\mathcal{B}_\lambda(\Phi, W_L, X, Y) = Y f_\lambda(\Phi, X) + \sum_{\alpha\beta AB} \epsilon_{\alpha\beta} W_{A\alpha L} W_{B\beta L} h_{\lambda AB}(\Phi, X). \quad (27.6.6)$$

(Lorentz invariance requires the spinor indices on the  $W_L$ s to be contracted with  $\epsilon_{\alpha\beta}$ .) The other symmetry is translation of  $X$  by an imaginary numerical constant,  $X \rightarrow X + i\xi$ , with  $\xi$  real. This changes the Lagrangian